

Flexo Spaces

SaaS-Level Booking System Roadmap

This roadmap breaks the Flexo Spaces booking infrastructure into manageable phases. The goal is to avoid feature explosion while still building a scalable and professional co-working booking platform.

Phase 1 — MVP Foundation

- Workspace CRUD
- Role-based authentication
- Booking schema
- Booking request flow
- Booking approval/rejection
- Basic owner dashboard
- Basic availability display

Phase 2 — Core Booking Engine

- Overlap detection logic
- Capacity calculations
- Remaining seat tracking
- Real-time availability
- Booking statuses
- Availability service
- Booking validation

Phase 3 — Payments & Reliability

- Razorpay integration
- Payment verification
- Booking locking system
- Pending request expiry
- Cancellation policies
- Refund engine
- Pricing snapshots

Phase 4 — Real-Time UX

- Socket.IO integration
- Live seat updates
- Availability badges
- Booking notifications
- Owner alerts
- Conflict warnings

Phase 5 — Smart Workspace Management

- Resource-based booking
- Meeting room booking
- Hot desk inventory
- Workspace impact analysis
- Conflict simulation
- Capacity optimization hints

Phase 6 — Advanced SaaS Features

- Negotiation workflow
- Recurring bookings
- Waitlists
- Trust scoring
- No-show tracking
- Audit logs

Phase 7 — Enterprise Scale

- Redis caching
- Background jobs
- Queue systems
- Analytics dashboard
- Occupancy reports
- Performance optimization

Recommended Timeline

Phase	Estimated Time
Phase 1	4–6 days
Phase 2	7–10 days
Phase 3	5–7 days
Phase 4	4–6 days
Phase 5	5–8 days
Phase 6	7–10 days
Phase 7	Ongoing optimization

Core Engineering Principles

- Never store infinite time slots
- Calculate availability dynamically
- Use overlap detection instead of isBooked booleans
- Snapshot pricing and workspace data
- Separate bookings from availability logic
- Use statuses to control workflows
- Build incrementally instead of solving everything at once

Flexo Spaces is no longer just a listing website. It is evolving into a real-time scheduling, inventory, and workspace management platform. The key to success is iterative development: build the core first, ship early, then improve based on discovered edge cases.